

Small Business Company Network **Design (Project 01)**



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Small Business Company Network Design

Plan Date: 02/26/2025 – 04/08/2025

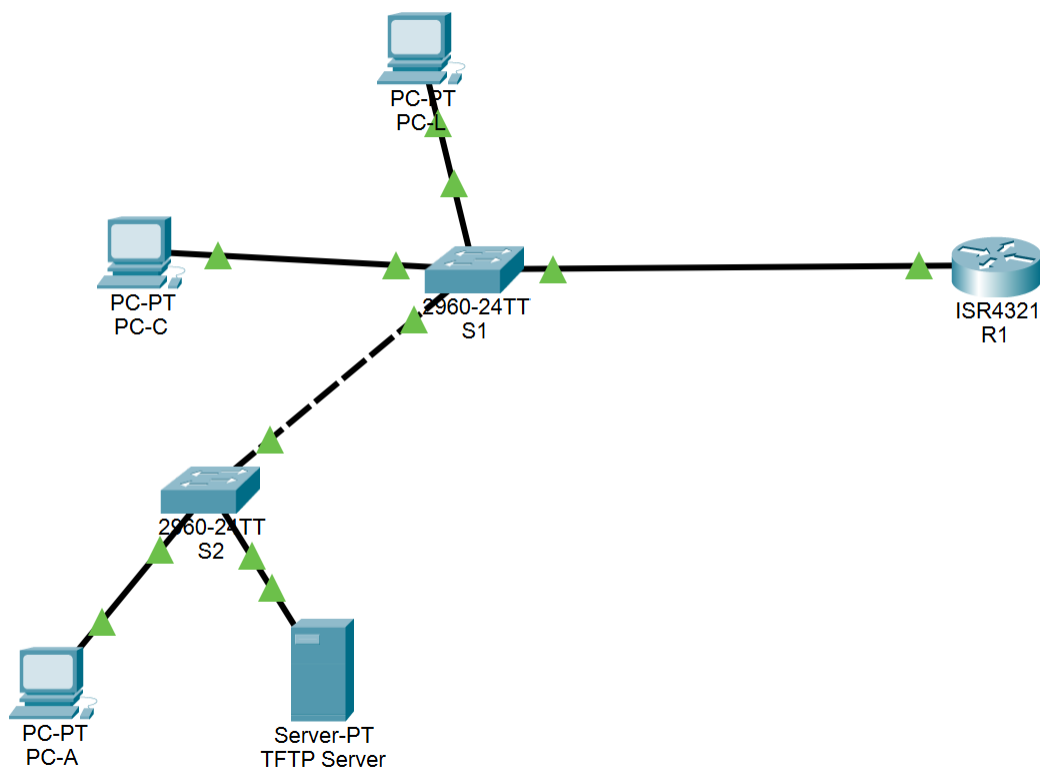
Company name: HSK Company

Network Engineer: Hashan Kodippilige

Description of Company: Providing Computer spare parts and support services.

List of Departments:	Management Department	03 Hosts
	Operations Department	09 Hosts
	Sales Department	08 Hosts

Topology Diagram:



Subnet List

Network Name	Number of Hosts	Network Mask	Network Address	1 st Address	Last Address	Broadcast Address
Management	03	255.255.255.240	192.168.1.0	192.168.1.1	192.168.1.14	192.168.1.15
Operations	09	255.255.255.240	192.168.1.16	192.168.1.17	192.168.1.30	192.168.1.31
Sales	08	255.255.255.240	192.168.1.32	192.168.1.33	192.168.1.46	192.168.1.47

Installation/Configuration

1. Install Wirshark

- a. Download 64-bit (For Windows 64-bit version) version file from www.wireshark.org to install in the PCs.
- b. Double clicks the file and start the installation process.
- c. Follow up the steps and if you already have a copy of Wireshark Click **Yes** to uninstall and previous version.
- d. In Wireshark Setup Wizard Click **Next**
- e. Click **I Agree** when the License Agreement Window displays.
- f. On the Choose component window, keep the default settings and Click **Next**
- g. Choose your desired shortcut options and click **Next**
- h. On the installation location window, keep the original location and click **Next**
- i. To capture live network data, WinPcap must be installed on your PC. If WinPcap is already installed on your PC, the Install check box will be unchecked. If you have an older version recommended that you allow the newer version to be installed by clicking the Install WinPcap x.x.x (version number) check box.
- j. Do not install USBPcap for normal traffic capture. Do Not Select the checkbox to install USBPcap (It is experimental).
- k. Wireshark starts installing its files and displays with the status of the installation. Click **Next** when Wireshark installation is complete and then click **Finish** to complete the process.
- l. In a separate window, accept the license agreement in the Npcap Setup Wizard if installing Npcap. Click **I Agree** to continue and then **Next** to finish the Npcap installation and click **Finish** to exit the Npcap installation.

2. Install Putty

- a. In the web browser, go to ninite.com
- b. Under the Development Tools, select Putty
- c. Click **Get your Ninite** button at the bottom of the window.
- d. After downloading, open the installer and run it.

3. Install TFTP server

- Go to www.solarwinds.com/free-tools/free-tftp-server
- Click **Download free tool**
- First Name: Hashan Last Name: Kodippilige
- Business Email: hashan@hsk.com
- Company: HSK Company
- Zip Code: 58203
- Business Phone: 7014108095
- Install
- Continue without adding
- Windows-Download now
- After you download, double click the file and start the installation process.

4. Saving configurations to TFTP server after each Session

4.1 Switches (Do it on each switch) and Router

Switches	Router (R1)
S1# copy flash:vlan.dat tftp Destination IP: 192.168.1.2 File name: s1-vlan.dat Save the configuration S1# copy running-config tftp Destination IP: 192.168.1.2 File name: s1-config Save the configuration S1# delete vlan.dat S1# erase startup-config	R1# copy running-config tftp Destination IP: 192.168.1.2 File name: r1-config Save the configuration R1# erase startup-config

5. Restoring configurations from TFTP server

5.1 Switches (Do it for each switch) and Router

Switches	Router (R1)
switch# copy tftp flash:vlan.dat (Answer questions) File name: s1-vlan.dat switch # copy tftp running-config (Answer questions) File name: s1-config Save the configuration	R1# tftp copy running-config tftp (Answer questions) File name: r1-config Save the configuration

Devices and Connections

Device Ports	Connected to Device Ports	Purpose
R1 G0/0/1	S1 F0/5	Trunk
S1 F0/1	S2 F0/1	Trunk
S2: F0/5-8	Management Group	VLAN 20
S1: F0/6-14	Operations Group	VLAN 30
S1: F0/15-22	Sales Group	VLAN 40

Basic Configurations for Network Devices (See Appendix -1)

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	172.24.72.1	255.255.255.248	N/A
	G0/0/1	N/A	N/A	
	G0/0/1.20	192.168.1.1	255.255.255.240	
	G0/0/1.30	192.168.1.17	255.255.255.240	
	G0/0/1.40	192.168.1.33	255.255.255.240	
	G0/0/1.1000	N/A	N/A	N/A
S1	VLAN 30	DHCP	255.255.255.240	192.168.1.1
S2	VLAN 20	DHCP	255.255.255.240	192.168.1.17
TFTP Server	NIC	192.168.1.2	255.255.255.240	192.168.1.1
PC-(A-B)	NIC	192.168.1.3 – 192.168.1.14	255.255.255.240	192.168.1.1
PC-(C-K)	NIC	192.168.1.18 – 192.168.1.30	255.255.255.240	192.168.1.17
PC-(L-S)	NIC	192.168.1.34 – 192.168.1.46	255.255.255.240	192.168.1.33

1. Router

- a. Assign a device name to the router.
- b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- c. Assign class as the privileged EXEC encrypted password.
- d. Assign cisco as the console password and enable login.
- e. Assign cisco as the VTY password and enable login.
- f. Encrypt the plaintext passwords.
- g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- h. Save the running configuration to the startup configuration file.

2. Switches

- a. Assign a device name to the switch.
- b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- c. Assign class as the privileged EXEC encrypted password.
- d. Assign cisco as the console password and enable login.
- e. Assign cisco as the VTY password and enable login.
- f. Encrypt the plain text passwords.
- g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- h. Save the running configuration to the startup configuration file.

3. PCs

- a. Assigning of PCs host address will be done through DHCP server and Assign TFTP server address as above table.

List of Each Feature Configured (See Appendix -2)

VLAN	Name	Interface Assigned
20	Management	S2: F0/6-8
30	Operations	S1: F0/6-14
40	Sales	S1: F0/15-22
999	ParkingLot	S1: F0/2-4, F0/23-24, G0/1-2 S2: F0/2-5, F0/9-24, G0/1-2
1000	Native	N/A

1. Create VLANs on both switches

- a. Create VLAN as shown in the table above
- b. Configure the management interface and default gateway on each switch using the IP address information in the Addressing Table.
- c. Assign used ports to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode.
- d. Assign all unused ports on the switch to the Parking Lot VLAN, configure them for static access mode, and administratively deactivate them.
- e. Verify configuration by issuing **Show Vlan Brief** command.

2. Configure Trunking

2.1 Manually configure trunk interface F0/1 on both Switches

- a. Change the switchport mode on interface F0/1 to force trunking. Do it on both switches.
- b. Set up Native vlan to 1000 on both switches.
- c. Specify that Vlan 20,30,40 and 1000 are allowed to cross the trunk.
- d. Verify configuration by issuing **Show interface trunk** command.

2.2 Manually configure S1's trunk interface F0/5

- a. Use the same configuration as F0/1 on switches to configure trunking on interface F0/5 on S1.
- b. Save configuration from Running to Startup.
- c. Verify configuration by issuing **Show interface trunk** command.

3. Configure Inter-Vlan Routing on Router

- a. Activate interface G0/0/1 on router.
- b. Configure Sub-interfaces for each VLAN as specified in the IP addressing table. All sub-interfaces use 802.1Q encapsulation. Do not configure IP address for Native VLAN. Include description for each sub-interface.
- c. Verify configuration by issuing **Show ip interface brief** command.

4. Configure Default Routing

- a. Configure G0/0/0 to IP address given on the table.
- b. Default route on Router to the ISP

5. Configure DHCP on Router (R1)

Exclude statically assigned addresses for all three vlans separately.

- a. Configure the pool name it with Vlan name and specify the subnet to use when assigning IP addresses.
- b. Configure dns server
- c. Configure the default router and domain name server for the network.
- d. Verify the DHCP configuration.

6. Configure R1 for PAT.

- a. Define the access list for internal network
- b. Configure the outside interface
- c. Configure inside interfaces
- d. Enable NAT overload (PAT)

7. Configure all network devices for basic SSH support.

- Create a local user with username SSHadmin and encrypted password @hsk123!
- Use hsk-spares.com as domain name.
- Generate crypto keys using a 1024-bit modulus.
- Configure the first five VTY lines on each device to support SSH connections only and to authenticate to local user database.

8. Verify the Network

From	Protocol	Destination	Result
PC-A	Ping	192.168.1.1	Success
PC-A	Ping	PC-C	Success
PC-A	Ping	PC-L	Success
PC-A	HTTPS	PC-L	Success
PC-A	HTTPS	PC-C	Success
PC-A	HTTPS	R1	Success
PC-A	SSH	PC-L	Success
PC-A	SSH	R1	Success
PC-C	Ping	192.168.1.17	Success
PC-C	Ping	PC-L	Success
PC-L	Ping	192.168.1.33	Success

9. Troubleshooting**- Configure wrong ip to the G0/0/0 interface**

I did configure ISP address to the G0/0/0 interface

- Configured g0/0/0 with *ip address 172.24.72.4 255.255.255.248*

- Internet is not working on all devices.

I did not configure DNS server on each DHCP pool, it prevents domain name resolution.

- Added *dns-server 8.8.8.8* to each DHCP pool for proper name resolution

ip nat inside command mistakenly configured to the physical G0/0/1 interface, which had no ip.

- Applied *ip nat inside* to each relevant sub interface G0/0/1.20, G0/0/1.30, G0/0/1.40, and G0/0/1.1000

Budget Estimation

Item	Unit Price (\$)	Quantity	Total Price (\$)
Router (ISR4321/K9)	1128.00	01	1128.00
Switch (WS-C2960-24TT-L)	625.00	02	1250.00
Pcs	749.99	20	14,999.80
Printer (Brother MFC-L3780CDW Color Multifunction Printer)	529.00	03	1587.00
Cables	5.13	50	256.50
Rack, Casings and Miscellaneous	-	-	2000.00
Total Amount			21,221.30

Appendix - 1

Basic Configurations for Network Devices

1. Router (R1)

- a. router(config)# hostname R1
- b. R1(config)# no ip domain lookup
- c. R1(config)# enable secret class
- d. R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
- e. R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
- f. R1(config)# service password-encryption
- g. R1(config)# banner motd \$ Unauthorized access is Prohibited! \$
- h. R1# copy running-config startup-config

2. Switches (This is for S1 and configure on both switches)

- a. switch(config)# hostname S1
- b. S1(config)# no ip domain lookup
- c. S1(config)# enable secret class
- d. S1(config)# line console 0
S1(config-line)# password cisco
S1(config-line)# login
- e. S1(config)# line vty 0 4
S1(config-line)# password cisco
S1(config-line)# login
- f. S1(config)# service password-encryption
- g. S1 (config)# banner motd \$ Unauthorized access is Prohibited! \$
- h. S1# copy running-config startup-config

Appendix - 2

List of Each Feature Configured

1. a. Create VLANs on both switches (This is for S1 and configure on both switches)

```
S1(config)# vlan 20
S1(config-vlan)# name Management
S1(config-vlan)# vlan 30
S1(config-vlan)# name Operations
S1(config-vlan)# vlan 40
S1(config-vlan)# name Sales
S1(config-vlan)# vlan 999
S1(config-vlan)# name ParkingLot
S1(config-vlan)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# exit
```

- b. S1(config)# interface vlan 30
S1(config-if)# ip address dhcp
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.1.17
S1(config)# end

```
S2(config)# interface vlan 20
S2(config-if)# ip address dhcp
S2(config-if)# no shutdown
S2(config-if)# exit
S2(config)# ip default-gateway 192.168.1.1
S2(config)# end
```

- c. S1(config)# interface range f0/6 - 14
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# exit
S1(config)# interface range f0/15 - 22
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 40
S1(config-if-range)# exit

```
S2(config)# interface range f0/6 - 8
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# exit
```

- d. S1(config)# interface range f0/2 - 4, f0/23 - 24, g0/1 - 2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# end

S2(config)# interface range f0/2 - 5, f0/9 - 24, g0/1 - 2
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 999
S2(config-if-range)# shutdown
S2(config-if-range)# end
- e. S1# show vlan brief (**This is for S1 and check it on both switches**)

2. Configure Trunking

2.1 Manually configure trunk interfaces F0/1 on both Switches (This is for S1 and configure on S2 also)

- a. S1(config)# interface f0/1
S1(config-if)# switchport mode trunk
- b. S1(config-if)# switchport trunk native vlan 1000
- c. S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
- d. S1# show interfaces trunk

2.2 Manually configure S1's trunk interface F0/5

- a. S1(config)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
- b. S1# copy running-config startup-config
- c. S1# show interfaces trunk

3. Configure Inter-Vlan Routing on Router (R1)

- a. R1(config)# interface g0/0/1
R1(config-if)# no shutdown
- b. R1(config)# interface g0/0/1.20
R1(config-subif)# description Management Network
R1(config-subif)# encapsulation dot1q 20

```
R1(config-subif)# ip address 192.168.1.1 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# description Operations Network
R1(config-subif)# ip address 192.168.1.17 255.255.255. 240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.40
R1(config-subif)# encapsulation dot1q 40
R1(config-subif)# description Sales Network
R1(config-subif)# ip address 192.168.1.33 255.255.255. 240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.1000
R1(config-subif)# encapsulation dot1q 1000 Native
R1(config-subif)# description Native Vlan
```

c. R1# show ip interface brief

4. Configure Default Routing

- a. R1(config)# interface g0/0/0
R1(config-if)# ip address 172.24.72.4 255.255.255.248
R1(config-if)# no shutdown
R1(config-if)# exit
- b. R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 172.24.72.1

5. Configure DHCP on Router (R1)

- a. R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.2
- b. R1(config)#ip dhcp pool Management
R1(config-dhcp)# network 192.168.1.0 255.255.255.240
- c. R1(config-dhcp)# default-router 192.168.1.1
R1(config-dhcp)# domain-name hsk-spare.com
- d. R1#show ip dhcp binding

Same step to create other configurations for Vlan 30 and Vlan 40

```
R1(config)#ip dhcp excluded-address 192.168.1.17
R1(config)#ip dhcp pool Operations
R1(config-dhcp)# network 192.168.1.16 255.255.255.240
R1(config-dhcp)# default-router 192.168.1.17
R1(config-dhcp)# domain-name hsk-spare.com
```

```
R1(config-dhcp)# end
R1#show ip dhcp binding

R1(config)#ip dhcp excluded-address 192.168.1.33
R1(config)#ip dhcp pool Sales
R1(config-dhcp)# network 192.168.1.33 255.255.255.240
R1(config-dhcp)# default-router 192.168.1.33
R1(config-dhcp)# domain-name hsk-spare.com
R1(config-dhcp)# end
R1#show ip dhcp binding
```

6. Configure R1 for PAT

- a. R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
- b. R1(config)# interface g0/0/0
R1(config-if)# ip nat outside
- c. R1(config)# interface g0/0/1.20
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.30
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.40
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.1000
R1(config-if)# ip nat inside
- c. R1(config)# ip nat inside source list 1 interface g0/0/0 overload

7. Configure all network devices for basic SSH support.

7.1 Configure all network devices for basic SSH support. (This is for R1, do this same step on both switches)

- a. R1(config)# username SSHadmin secret \$hsk123!
- b. R1(config)# ip domain name hsk-spare.com
- c. R1(config)# crypto key generate rsa general-keys modulus 1024
- d. R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit

Appendix – 3
Each Device Configuration

Router (R1)

```
router(config)# hostname R1
R1(config)# no ip domain lookup
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config)# service password-encryption
R1(config)# banner motd $ Unauthorized access is Prohibited! $
R1# copy running-config startup-config
```

```
R1(config)# interface g0/0/1
R1(config-if)# no shutdown
```

```
R1(config)# interface g0/0/1.20
R1(config-subif)# description Management Network
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.1.1 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# description Operations Network
R1(config-subif)# ip address 192.168.1.17 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.40
R1(config-subif)# encapsulation dot1q 40
R1(config-subif)# description Sales Network
R1(config-subif)# ip address 192.168.1.33 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.1000
R1(config-subif)# encapsulation dot1q 1000 Native
R1(config-subif)# description Native Vlan
R1# show ip interface brief
```

```
R1(config)# interface g0/0/0
R1(config-if)# ip address 172.24.72.4 255.255.255.248
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 172.24.72.1
```

```
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.2
R1(config)#ip dhcp pool Management
R1(config-dhcp)# network 192.168.1.0 255.255.255.240
R1(config-dhcp)# default-router 192.168.1.1
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# domain-name hsk-spare.com
R1#show ip dhcp binding
```

```
R1(config)#ip dhcp excluded-address 192.168.1.17
R1(config)#ip dhcp pool Operations
R1(config-dhcp)# network 192.168.1.16 255.255.255.240
R1(config-dhcp)# default-router 192.168.1.17
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# domain-name hsk-spare.com
R1(config-dhcp)# end
```

```
R1(config)#ip dhcp excluded-address 192.168.1.33
R1(config)#ip dhcp pool Sales
R1(config-dhcp)# network 192.168.1.33 255.255.255.240
R1(config-dhcp)# default-router 192.168.1.33
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# domain-name hsk-spare.com
R1(config-dhcp)# end
R1#show ip dhcp binding
```

```
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)# interface g0/0/0
R1(config-if)# ip nat outside
```

```
R1(config)# interface g0/0/1.20
R1(config-if)# ip nat inside
```

```
R1(config)# interface g0/0/1.30
R1(config-if)# ip nat inside
```

```
R1(config)# interface g0/0/1.40
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.1000
R1(config-if)# ip nat inside

R1(config)# ip nat inside source list 1 interface g0/0/0 overload

R1(config)# username SSHadmin secret $hsk123!
R1(config)# ip domain name hsk-spare.com
R1(config)# crypto key generate rsa general-keys modulus 1024
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit

R1# copy running-config startup-config
```

Switch (S1)

```
switch(config)# hostname S1
S1(config)# no ip domain lookup
S1(config)# enable secret class
S1(config)# line console 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config)# line vty 0 4
S1(config-line)# password cisco
S1(config-line)# login
S1(config)# service password-encryption
S1 (config)# banner motd $ Unauthorized access is Prohibited! $
S1 (config)# exit
S1# copy running-config startup-config

S1(config)# vlan 20
S1(config-vlan)# name Management
S1(config-vlan)# vlan 30
S1(config-vlan)# name Operations
S1(config-vlan)# vlan 40
S1(config-vlan)# name Sales
S1(config-vlan)# vlan 999
```

```
S1(config-vlan)# name ParkingLot
S1(config-vlan)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# exit
```

```
S1(config)# interface vlan 30
S1(config-if)# ip address dhcp
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.1.17
S1(config)# interface range f0/6 - 14
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# exit
S1(config)# interface range f0/15 - 22
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 40
S1(config-if-range)# exit
```

```
S1(config)# interface range f0/2 - 4, f0/23 - 24, g0/1 - 2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# end
```

```
S1# show vlan brief
```

```
S1(config)# interface f0/1
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# switchport trunk native vlan 1000
S1(config-if-range)# switchport trunk allowed vlan 20,30,40,1000
S1(config-if-range)# end
S1# show interfaces trunk
```

```
S1(config)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
S1# copy running-config startup-config
S1# show interfaces trunk
```

```
S1(config)# username SSHadmin secret $hsk123!  
S1(config)# ip domain name hsk-spare.com  
S1(config)# crypto key generate rsa general-keys modulus 1024  
S1(config)# line vty 0 15
```

```
S1(config-line)# transport input ssh  
S1(config-line)# login local  
S1(config-line)# exit  
S1(config)# exit
```

```
S1# copy running-config startup-config
```

Switch (S2)

```
switch(config)# hostname S2  
S2(config)# no ip domain lookup  
S2(config)# enable secret class  
S2(config)# line console 0  
S2(config-line)# password cisco  
S2(config-line)# login  
S2(config)# line vty 0 4  
S2(config-line)# password cisco  
S2(config-line)# login  
S2(config)# service password-encryption  
S2 (config)# banner motd $ Unauthorized access is Prohibited! $  
S2 (config)# exit  
S2# copy running-config startup-config
```

```
S2(config)# vlan 20  
S2(config-vlan)# name Management  
S2(config-vlan)# vlan 30  
S2(config-vlan)# name Operations  
S2(config-vlan)# vlan 40  
S2(config-vlan)# name Sales  
S2(config-vlan)# vlan 999  
S2(config-vlan)# name ParkingLot  
S2(config-vlan)# vlan 1000  
S2(config-vlan)# name Native  
S2(config-vlan)# exit
```

```
S2(config)# interface vlan 20
S2(config-if)# ip address dhcp
S2(config-if)# no shutdown
S2(config-if)# exit
S2(config)# ip default-gateway 192.168.1.1
S2(config)# end
```

```
S2(config)# interface range f0/6 - 8
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# exit
```

```
S2(config)# interface range f0/2 - 5, f0/9 - 24, g0/1 - 2
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 999
S2(config-if-range)# shutdown
S2(config-if-range)# end
```

```
S2# show vlan brief
```

```
S2(config)# interface f0/1
S2(config-if-range)# switchport mode trunk
S2(config-if-range)# switchport trunk native vlan 1000
S2(config-if-range)# switchport trunk allowed vlan 20,30,40,1000
S2(config-if-range)# end
S2# show interfaces trunk
```

```
S2(config)# username SSHadmin secret $hsk123!
S2(config)# ip domain name hsk-spare.com
S2(config)# crypto key generate rsa general-keys modulus 1024
S2(config)# line vty 0 4
S2(config-line)# transport input ssh
S2(config-line)# login local
S2(config-line)# exit
S2(config)# exit
```

```
S2# copy running-config startup-config
```